Placement Portal Application – V2

Project Report

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1. Student Details

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2. Project Details

Project Title: Placement Portal Application – V2

Problem:
Institutes need a centralized system to manage placement activities such as company registration, placement drives, student applications, and recruitment tracking. Manual methods like spreadsheets and emails are inefficient and difficult to maintain.

Objective:
Develop a **web-based placement management system** that connects **Admin, Companies, and Students** through a single platform.

Technology Approach:
A **full-stack architecture** was used:

- **Frontend:** VueJS + Bootstrap
- **Backend:** Flask REST APIs
- **Database:** SQLite with SQLAlchemy ORM
- **Authentication:** JWT (Flask-JWT-Extended)

Frontend communicates with backend using **JSON API requests**.

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3. System Roles and Features

Admin

- Approve / reject company registrations
- View and manage students and companies
- Activate / block student accounts
- View placement drives
- View all applications

Company

- Register company profile
- Login after admin approval
- Create placement drives
- View student applications
- Update application status

Student

- Register profile
- Upload resume
- View approved placement drives
- Apply for drives
- Track application status

System Rules

- Only **approved companies** can create drives
- Only **approved drives** are visible to students
- **Duplicate applications are prevented**
- Admin can **activate or block accounts**

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4. AI / LLM Declaration

AI tools were used only for **learning assistance and debugging**.

Usage included:

- Understanding Flask APIs
- Learning VueJS components

- Debugging errors
- Improving documentation

Core system design, database schema, and integration were **implemented manually**.

Estimated AI assistance: **~20–25% for development support.**

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5. Frameworks and Libraries

Backend

- Flask
- Flask-JWT-Extended
- Flask-CORS
- SQLAlchemy
- Werkzeug Security
- SQLite

Frontend

- VueJS
- Bootstrap

Other Tools

- JSON APIs for communication
- File upload system for resumes

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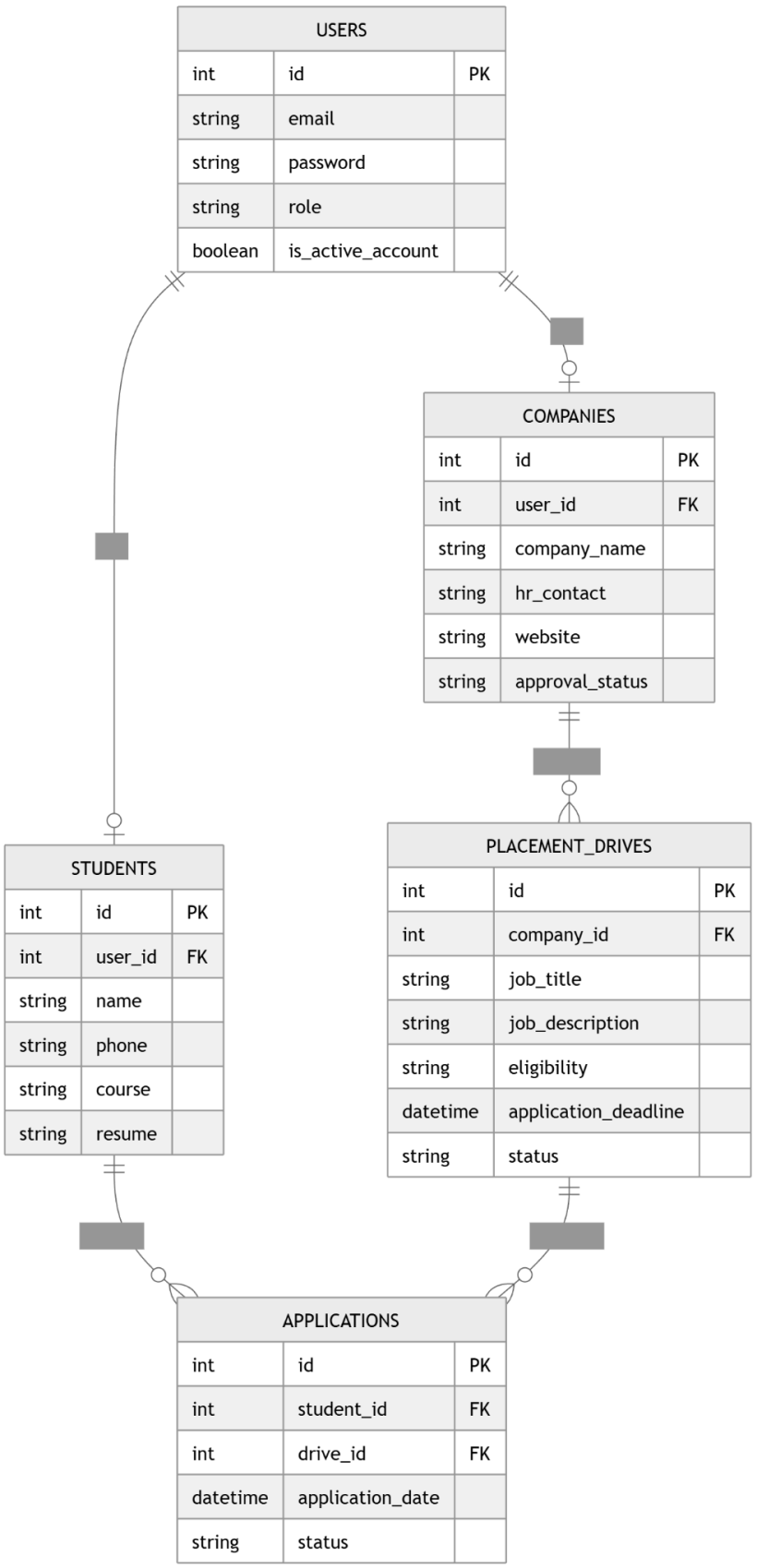
6. Database ER Design



7. API Endpoints

Authentication

POST /api/login

Student APIs

POST /api/register/student
GET /api/student/me
GET /api/student/available-drives
GET /api/student/my-applications
POST /api/student/apply/<drive_id>
PUT /api/student/profile/update

Company APIs

POST /api/register/company
GET /api/company/me
GET /api/company/stats
GET /api/company/drives
POST /api/company/drives
GET /api/company/applications
PUT /api/company/applications/<id>/status

Admin APIs

GET /api/admin/dashboard
GET /api/admin/students
GET /api/admin/students/search
GET /api/admin/student/<id>
PUT /api/admin/student/<id>/block
PUT /api/admin/student/<id>/activate
DELETE /api/admin/student/<id>

GET /api/admin/companies
PUT /api/admin/company/<id>/status

GET /api/admin/drives
PUT /api/admin/drive/<id>/status

GET /api/admin/applications

Presentation Video

Presentation Video:

https://drive.google.com/file/d/1u8yR8p_KGyI0y018c7ee3A2MqD1vb5f8/view